Developed a high-fidelity Dissolved Oxygen (DO) forecasting model using 17,965 hourly data points from the Han River (2022–2024). The PatchTST transformet architecture outperformed other models, capturing complex daily periodicities and long-term trends with an R^2 of 0.970 on unseen future data.

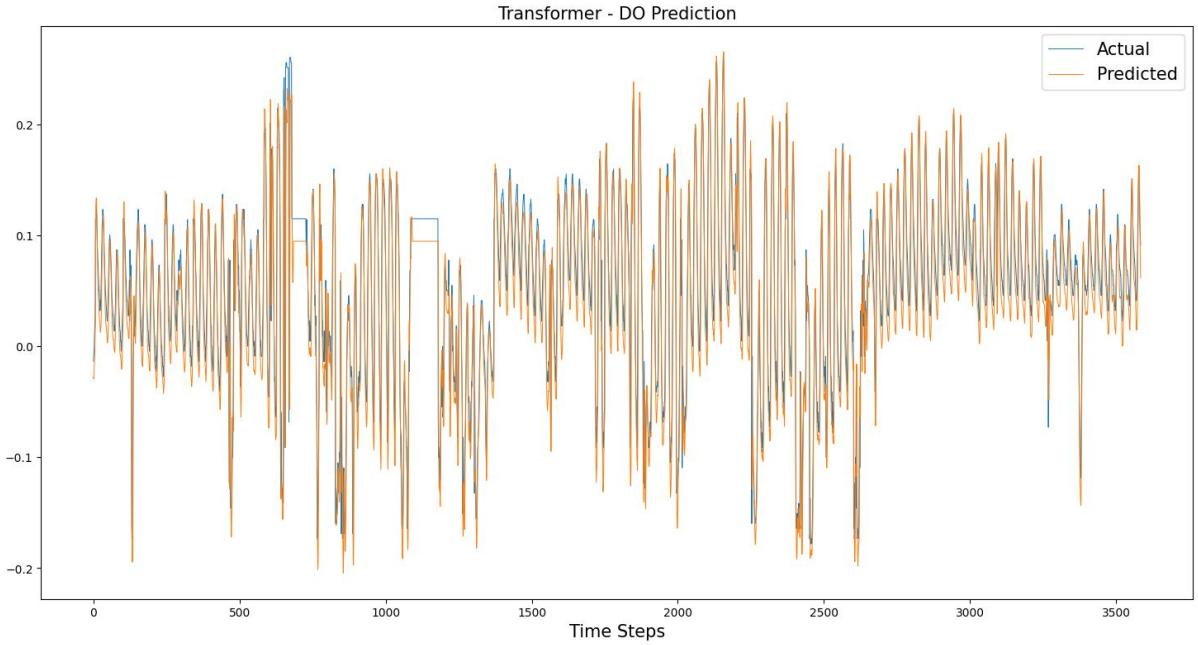

Figure 1: Transformer Model Validation on Time-Series Test Dataset

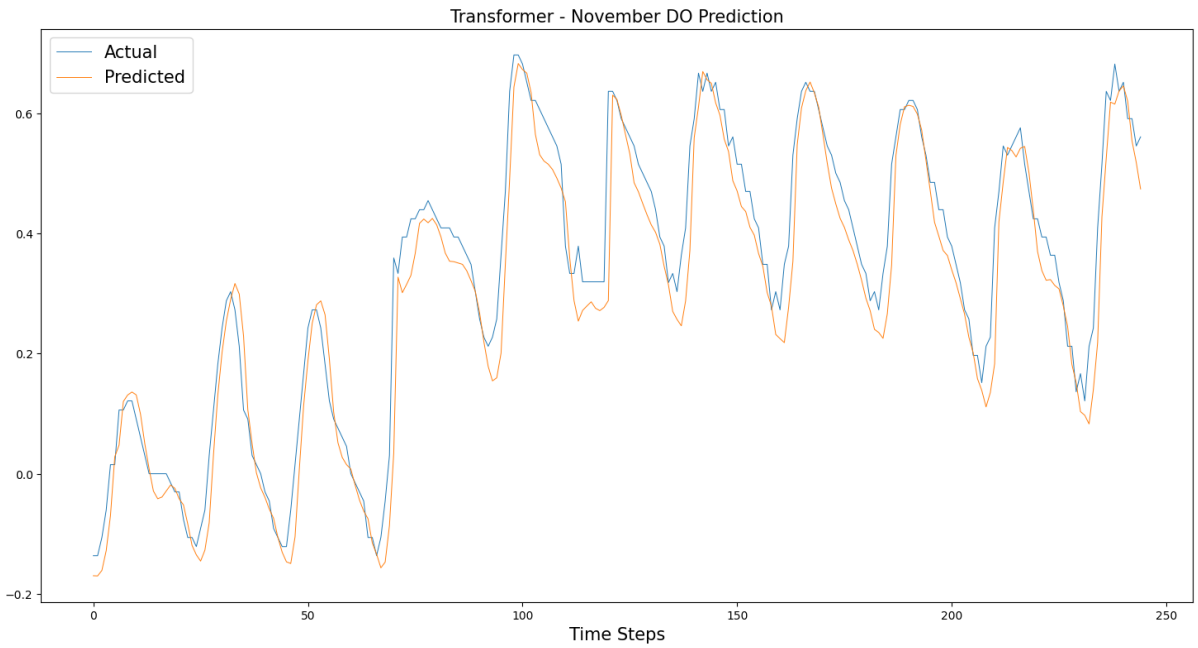

Figure 2: Out-of-Sample Forecasting for Unseen Data (November 2024)

Table 1: Comparative Analysis of Performance Metrics for Deep Learning Architectures on Unseen Data

Model	MSE	RMSE	MAE	R^2
LSTM	0.000	0.017	0.012	0.961
GRU	0.001	0.029	0.015	0.915
Transformer	0.000	0.015	0.010	0.970